

Progress Summary

Completed Items:

- Read suggested topic papers:
 - <https://doi.org/10.1145/2213556.2213582>
 - <https://doi.org/10.1145/3299869.3300078>
 - <https://doi.org/10.1145/3572751.3572757>
- Decide main paper/implementation of our paper: [Towards Model-based Pricing for Machine Learning in a Data Marketplace](#)
- Research additional papers and related works for background information
- Read [Towards Model-based Pricing for Machine Learning in a Data Marketplace](#), [Data Pricing in Machine Learning Pipelines](#), and [Data Shapley: Equitable Valuation of Data for Machine Learning](#)
- Determine direction of project / the purpose of our paper, formalize a research question
- Conduct research of decided papers and compile notes / quotes
- Outline our paper with the organized notes

Action Items:

- ☐ Complete initial draft of the paper (split up sections)
- ☐ Refine draft collaboratively
- ☐ Finalize paper
- ☐ Create slides for in-class presentation

Practical Model-Based Pricing for Real-World Data Marketplaces

ABSTRACT

mbp paper is great, but it is not applicable to the market
the limitations we focus on / address

- multi-seller market revenue allocation (shapley)
- not able to apply to evolving models / continual learning models, which are becoming more common and important
- not efficient – takes a lot of time to run for larger models (which are more realistic in the real-world)

With the rapid expansion of AI systems, pricing data for machine learning is a complex yet increasingly important problem. One solution is to price not the raw data itself, but the outputs of trained models, an approach known as model-based pricing (MBP). This is done by adjusting model accuracy through controlled Gaussian noise injection, enabling sellers to maximize revenue while making data more accessible. However, this approach remains limited in its applicability to real-world data marketplaces. In practice, markets often involve multiple data creators or contributors (sellers) and models are typically non-convex and evolving rather than static. As such, in this work, we extend the MBP framework to better reflect realistic marketplaces by implementing Shapley-value revenue allocation, exploring non-convex models, and improving efficiency. Our findings show that...

1. INTRODUCTION

why does data pricing matter?

- “Data analytics using machine learning (ML) has become ubiquitous in science, business intelligence, journalism and many other domains.”
- “Data is the fuel powering artificial intelligence and therefore it has value.”
- “Various sectors such as health-care and advertising are becoming more and more dependent on the data generated by individuals; a dependence similar to how they depend on labor and capital.”
- Machine learning depends on data, which implies that data has economic value that must be priced accordingly

what makes ML data pricing difficult?

- Data has context-dependent value, which makes it difficult to price in practice
 - “The value of each datum depend on the learning algorithm, the performance metric as well as on the rest of the train data.”

- “Assigning value to data in the setting of supervised learning is not equal to assigning a universal value to it.”
- Since data in machine learning contexts do not contribute equally, revenue allocation is important to ensure fair and representative pricing
 - “All data sources are not created equal.”
 - “Assigning value to data... is not equal to assigning a universal value to it.”
- In machine learning, data value is contextual and not all data contributes equally, making the question of equitable pricing incredibly complex
 - reframes as a data valuation problem over market problem → easier to tie in shapley as core support

how has data been priced thus far and what are the limitations with it?

- However, current data markets are inefficient and unaffordable to those looking for specific use-cases rather than entire datasets
 - “Existing pricing schemes either force users to buy the whole dataset or support simplistic pricing mechanisms, without any awareness of the ML task downstream.”
 - “This means that valuable datasets may not be affordable to potential buyers with limited budgets.”
- Furthermore, realistic machine learning markets are multi-agent, meaning multiple components go into training and enhancing models
 - “MLDM is a distributed ML platform where data is traded among agents.”
 - “Individual agents continuously gather data to enhance their learning models.”
 - This makes a lot of existing solutions (like MBP) unrealistic and impractical

introduce model-based pricing as a potential solution

- Model-based pricing is a solution that reframes the pricing problem
 - “Instead of selling raw data to the buyers, the market can directly sell ML model instances with different accuracy options.”
 - “a simple and efficient noise-injection mechanism that realizes an MBP framework with formal guarantees. Specifically, the broker first trains the optimal model instance, which is a one-time cost. When a buyer requests a model instance, the broker adds random Gaussian noise to the optimal model and returns it to the buyer. Our proposed mechanism avoids training a model instance from scratch and is able to achieve realtime interaction.”

discuss limitations of model-based pricing

- However, the MBP framework does not accurately reflect real data markets
 - “The development of robust machine learning models... frequently involves collaboration between multiple entities.”

- “Non-convex models such as neural networks are out of scope.”
 - “We show that the optimization is a computationally hard problem.”
- Model-based pricing is a step towards equitably pricing in machine learning data markets, but before the framework can be applied in practice there must be further theoretical support to address the limitations presented in the paper

what we are trying to accomplish with our paper

- We extend upon model-based pricing to be more practical for multi-agent machine learning data markets by integrating Shapley value for fair revenue allocation.
- address shortcomings of mbp
 - Use shapley
 - Issues with valuing nonconvex ml models with MBP
 - Takes a lot of time to run for bigger models

2. RELATED WORKS

Because pricing data has been an important research question for years, there is a lot of content to discuss. We will focus on describing the state of the real world data marketplace, how raw data is priced, machine learning as a service (MLaaS), and revenue allocation for multiple data creators or contributors.

2.1 Real World Data Marketplace

To provide background context on real-world data marketplaces, we will discuss data marketplaces, pricing strategies and market types, and desiderata.

- Data Marketplaces
- Pricing Strategies and Market Types (discuss these sections jointly - not too long)
- Desiderata of Data Pricing
- “Agarwal et al. [1] summarize five properties making data a unique asset. First, data can be replicated at zero marginal cost. Second, the value of data is inherently combinatorial. Third, the value of data varies widely among different buyers. Last, the usefulness of data lies in the value of information derived from it, which is difficult to verify a priori.”

2.2 Pricing Raw Data

Focusing on the intrinsic value of a dataset is one way that raw datasets are priced.

Another way to price raw datasets is crowdsensing data, which is focusing on truthful reporting of collection costs. This is difficult because motivating data labelers to accurately label data is an ongoing issue—one that people attempt to solve with...

Query-based pricing depends on the amount of information revealed or uncertainty reduction.

Privacy compensation focuses on the magnitude of injected noise and privacy loss.

- **Computational Hardness:** Determining versions and prices to maximize revenue is NP-hard, forcing the use of heuristic algorithms that may not find the absolute optimal solution.
- **Arbitrage Loops:** In query-based pricing, less carefully designed models can lead to arbitrage loopholes, allowing buyers to reconstruct expensive data results from cheaper query bundles.
- **History-Awareness Costs:** In systems that prevent charging users twice for the same data, the seller is burdened with providing reliable storage to maintain the query history of every user.
- **Privacy-Accuracy Trade-offs:** In personal data marketplaces, a significant limitation is that conservative individuals may refuse to participate entirely if the privacy loss exceeds their personal threshold, potentially reducing the overall accuracy of the resulting data product.
- “A limitation of the discussed pricing models is that data sets are priced without considering their down-stream applications. Fernandez et al. [30] argue that the value of a data set to customers is usually task dependent and cannot be evaluated by the intrinsic properties of the data set alone. As the pricing models of raw data sets are agnostic to the down-stream applications of raw data sets, these pricing models can be used in machine learning pipelines of building both supervised and unsupervised machine learning models.”

2.3 Machine Learning as a Service

- MBP: “Unfortunately, their pricing model only works for machine learning models trained with strictly convex objective functions.”
- 6.2 Pricing Raw Data Products Versus Machine Learning Models

2.4 Revenue Allocation

- Assign numerical worth to training data, quantifying each data point's contribution to a model's performance. These frameworks, such as Shapley value, Data Banzhaf, or Reinforcement Learning (DVRL), identify high-value data, detect noisy/corrupted samples, improve data quality, and enable fair compensation in collaborative AI.
 - Data Banzhaf: <https://arxiv.org/abs/2205.15466>
 - [Data Valuation using Reinforcement Learning](#)
 - “Yoon et al. [106] propose a reinforcement learning algorithm to value data points. They learn a data value estimator that estimates data values and selects the most valuable samples to train a target classifier. They

jointly learn the data value estimator and the corresponding classifier, which enables the classifier and the data value estimator to improve the performance of each other. However, this method cannot guarantee fair revenue distribution among participants

- Section 4: Most of the existing revenue allocation methods are developed in the settings that supervised machine learning models are jointly trained. The participants are rewarded based on the contributions of their data sets to the utility of the jointly trained machine learning model. To adapt existing pricing models to scenarios where unsupervised machine learning models are jointly trained, the major challenge is to develop a utility function that participants can all agree on. For some traditional unsupervised machine learning models, there are some widely accepted performance metrics that can serve as the utility functions. For example, Silhouette Coefficient [86] and Calinski-Harabasz index [12] are widely used to evaluate the performance of clustering algorithms when groundtruth clusters are unknown. However, developing a utility function for some unsupervised models, such as pre-trained deep language models [26, 9], may be challenging, as they are evaluated differently in many down-stream machine learning tasks.

3. METHODS

intro

3.1 Revenue Allocation Using Shapley Value

intro: what is shapley and what is the limitation of mbp that we are targeting here

- Shapley is used not to determine the price of model outputs, but to determine how revenue derived from those outputs should be allocated across contributors in a multi-agent data marketplace
- This directly addresses a limitation of MBP, which treats the seller as a unified entity
- As such, the purpose of integrating Shapley into the MBP framework is not to improve the pricing, but to determine revenue allocation to better reflect the multi-agent machine learning data market
 - we extend the mbp framework without breaking it; shapley pretty seamlessly fits into the framework yay!

how does shapley factor into mbp

- Model-based pricing determines value based on model outputs rather than raw data
 - “Instead of selling raw data to the buyers, the market can directly sell ML model instances with different accuracy options.”
- In doing so, the framework abstract market sellers as a cohesive unit, whereas in reality machine learning markets operate under multi-agent sellers

- shapley enables us to explore this abstract and turn it into something more applicable through revenue allocation, an equitable assignment of value based on relative contribution
- Since model-based pricing looks at pricing model performance and shapley allocates model performance, the two theories are compatible
 - “We view the supervised machine learning problem as a cooperative game... with the model performance as the utility.”

how shapley expands the market structure

- Model-based pricing has three components to the market structure, with the seller being an abstracted singular unit
 - “Our framework involves three types of agents that interact in the setting of a data marketplace: the seller, the broker and the buyer.”
- shapley treats contributors as players
- With Shapley, the MBP framework becomes more expansive and representative of real machine learning data market dynamics
- The seller node is now described as a shared-revenue model whose individual values are determined via Shapley

how to integrate shapley in practice

- Model-based pricing works with a trained model, so Shapley can be factored in prior to pricing by decomposing the value of the trained model
- As such, there is no significant adjustment that needs to be made to the pricing mechanism in the MBP framework
 - shapley operates on performance, so performance → mbp pricing → revenue
- create a diagram to illustrate the new flow of mbp with shapley factored in
 - train the model → compute contributions (performance) → allocate value → inject noise → price
- emphasize the difference with shapley – what changed?
 - only affects revenue distribution
 - does not affect the price-error curves or how the framework operates
 - tie back to our main argument: we are not trying to improve the pricing, but rather adjust mbp to make it more applicable in realistic settings

3.2 Practical Application for Non-Convex Models

Why does the original model only work with convex objective functions?

- The proposed way to sell models with different accuracy is as follows
 - Train optimal model with all data possible
 - Add gaussian noise with variance σ to weights.

- In the paper, they prove this gaussian noise affects the accuracy of the answer in several important ways
 - Expected error is a monotonic increasing function to noise variance

briefly mention

- why are non-convex ml models difficult to price?
 - Error monotonicity: perturbations in one weight do not lead to predictable outcome perturbations. could push models out of local minima into better areas, i.e. increasing variance does not necessarily return worse outcomes, and does not result in predictably bad outcomes on average either without knowing global behavior.
 - Lemma 3 contrapositive states that failure to show error monotonicity also results in failure to be arbitrage free in convex models, but in non-convex problems this may not be true
 - Buyers have no reason to expect arbitrage opportunities if seller gives weights in different local minima in non-convex

look into

- how can we adapt it to make it more practical?
 - The prior constraints have one major issue when applied to non-convex models; without knowledge of expected outcome given any combination of model weights, “worsening” the model is not predictable.
 - Imagine a seller wants a very good non-convex model (assuming something better than what is optimal in their allocated compute time) there are different ways of offering “worse” systems.
 - One possible solution is for sellers to charge buyers based on time trained, some curve of price to additional performance (search for x percent with y more dollars, but give up of gains dont materialize at a certain rate). For the seller, they can cache predictive outcome values for future buyers. Future buyers would either have the opportunity to retrain from the previous model or start at the newest model with their own performance curve.
 - Conditions that give tractability in this pricing may also lead to tractability in the solutions for non-convex ml applications, vice versa.
- It seems more likely are pessimistic results rather than positive ones. Perhaps this is one of the reasons that there doesnt exist much progress in the non-convex regimes.

4. FUTURE WORK

what are the remaining limitations?

Limitations: the setting of supervised learning with a fixed hypothesis space using a fixed set of features and a strictly convex model objective.

BIG PROBLEMS:

- can't mix and match features: buyer can cheat
- does the broker have the model that you want for a specific dataset?

Future directions

- First, more complex ML models such as Bayesian networks and artificial neural networks, are also frequently used.
- Nonrelational data (images, text, etc.) might require complex feature extraction, possibly implicitly within an ML model (as in deep learning [24]).
- Second, we assumed that buyers know which ML model/hypothesis they want. This is a reasonable starting point because most existing cloud ML platforms assume the buyer picks the ML model. But in practice, users often perform model selection and explore different ML models [12, 33] and refine their choices iteratively [22]. Handling complex models and feature/model selections can be challenging, as it may require a revised arbitrage-freeness notion based on probabilistic guarantee and is left to future work.
- Third, in many cases data comes with privacy constraints, and therefore integrating model-based pricing with data privacy is also a core future challenge.
- Finally, more complicated buyer models as well as trade-offs between revenue and fairness can be further explored in the revenue optimization.

5. CONCLUSION

The MBP in chen et al is interesting proposal for a pricing model and if applicable to industrial data providers promises a useful theoretically sound pricing metric to base their operations off of. Given the lack of industrial adoption and the issues we point out in this paper, the limitations unfortunately currently outweigh the benefits for industrial purposes. Important variables such as multi-agent sellers, computational constraints given large datasets on even just convex models, as well as the lack of applicability to non-convex settings leave the pricing model in a precarious position in regards to adoptability.

REFERENCES

1. [Towards Model-based Pricing for Machine Learning in a Data Marketplace](#)
2. [What is your data worth? Equitable Valuation of Data](#)
3. [Shapley value-based data valuation for machine learning data markets](#)
4. [Data Pricing in Machine Learning Pipelines](#)